

XIONGXIANG CHENG

Shenyang, Liaoning - 110819, China
(+86) 19848083415 | 20234388@stu.neu.edu.cn

EDUCATION

Northeastern University (China, Project 985)	Sep.2023 - Jun.2027 (Expected)
- Materials Science and Engineering (B.A.) Course: Properties of Materials(4/92), Dynamics(98), Intelligent Energy Storage Systems(94)	GPA:86.034/100 - Rank:15%
Université de Toulouse (France)	Sep.2023 - Jun.2027 (Expected)
- ELECTRONIQUE, ENERGIE ELECTRIQUE, AUTOMATIQUE (B.A.) Course: Digital Data Processing(96), Nonlinear Dynamical Systems(90), Mathematics	

EXPERIENCE

Research on Near-Optimal Control of Nonlinear Systems Based on SVR	
Project Leader / Supervisor: Prof. Hanguang Su Core Contribution: Proposed a Value Iteration-based near-optimal control scheme specifically targeting continuous-time affine nonlinear systems. Innovatively integrating SVR into the ADP framework to replace neural networks, effectively overcoming local optima. Technical Implementation: Enabling online optimal decision-making for partially unknown dynamics via Integral Reinforcement Learning without initial stabilizing policies. Rigorously proving continuous-time Value Iteration convergence and SVR weight stability via convex optimization theory.	2024.02-Present
Project on High-Quality Development Special Project No.179, MIIT	
Assistant / Supervisor: Prof. Hanguang Su - Collaborating with project team members to collect the necessary information and data. - Contributed to the optimal control research for the drafting of three invention patents related to wafer defect detection algorithms.	2024.08-2026.07
Research on Multi-tier Energy Storage Collaborative Control Device Manufacturing	
Member / Supervisor: Prof. Hanguang Su Core Contribution: Developed the DC Motor and Transformer Parameter Calculator (Software Copyright: 2025SR0345659) to accelerate the hardware calibration process. - Integrated Long Short-Term Memory networks into the ADP framework to predict load fluctuations and optimize real-time energy distribution.	2024.12-2025.12

PUBLICATIONS

Software Copyright: <i>Parameter Calculator for DC Motors and Transformers (DC-TCalc) V1.0</i>	
Second Author, Granted, Reg. No.: 2025SR0345659	2025.02
Patent on Invention: <i>A Support Vector Regression-Based Control Approach for Approximate Dynamic Programming</i>	
First Author, Patent Granted, Patent Application No.: 202510637337.1	2025.08

AWARDS AND SCHOLARSHIPS

Second-class Scholarship, Northeastern University	2024.11
Outstanding Student, Northeastern University	2024.12
2025 National Best Team Award (Top 10 teams), Cloud Teaching of Zhixing Program	2025.11
First-class Scholarship (Top 1 of Major), Northeastern University	2025.11
Outstanding Student Pacesetter (Top 1 of Major), Northeastern University	2025.12
Weichai Power Scholarship (Top 1 of Major), Northeastern University	2025.12
Honorable Mention , 2026 Mathematical Contest In Modeling	2026.05
French Government Scholarship (6 recipients nationwide), Embassy of France in China	2026.05

SKILLS

Languages: Mandarin (native), English (IELTS 6.5), French (TCF B1).
Skills: LTspice, Matlab, SPSS, Python, CATIA etc.